

Xinsong Feng

(+1) 213-791-1673
xsfeng@gatech.edu
xfeng01.github.io

Research Interests

My research spans reinforcement learning, generative models, and language models. I am currently especially interested in diffusion language models, particularly in understanding their behavior and improving inference efficiency and performance. I aim to develop AI systems that are trustworthy, interpretable, and theoretically grounded.

Education

Georgia Institute of Technology
Ph.D. in Computer Science

Atlanta, USA
Aug. 2026–Present

University of California, Los Angeles
M.S. in Electrical and Computer Engineering, GPA: 4.00/4.00

Los Angeles, USA
Sep. 2023–Jun. 2025

Chongqing University
B.Eng. in Communication Engineering, Major GPA: 3.94/4.00

Chongqing, China
Sep. 2019–Jun. 2023

Research Experience

Data-Driven Decision Intelligence (D3I) Lab, William & Mary
Research Assistant

Aug. 2025–May. 2026
Advisor: Haipeng Chen, Assistant Professor

DELA: Learning Controllable Edits for Diffusion Language Models

Oct. 2025–Jun. 2026

- Developed DELA, a plug-in framework that improves frozen diffusion language models through explicit draft-level editing actions: INSERT, DELETE, REMASK, and STOP.
- Designed editor and selector policies for proposing and choosing beneficial sequence edits, separating structural control from token-level denoising.
- Designed a curriculum learning strategy that combines inverse-edit supervision with rollout-guided preference learning for effective edit proposal and selection.
- Built the full experimental pipeline for candidate generation, verifier-based selection, and benchmark evaluation under a fixed-backbone setting.

Constrained Optimization via RL and ODE-based Models

Mar. 2025–Sep. 2025

- Proposed Constrained Markov Flow Optimizer (CMFO), a framework unifying reinforcement learning and ODE-based generative models for constrained optimization.
- Formulated constrained optimization as sequential decision-making over iterative refinement dynamics, establishing a connection between policy learning and ODE-based optimization.
- Introduced RelaxCompletion for differentiable feasibility handling and achieved state-of-the-art performance with substantially faster inference than classical solvers.

University of California, Los Angeles
M.S. Student Researcher

Sep. 2023–Jun. 2025
Research collaboration with Prof. Haipeng Chen

Offline Reinforcement Learning with Generative Trajectory Policies

Aug. 2024–May 2025

- Proposed Generative Trajectory Policies (GTP), a unified offline RL framework connecting diffusion, flow matching, and consistency models through a continuous-time trajectory formulation.
- Derived a closed-form score approximation from offline data, eliminating iterative ODE solvers and enabling stable and efficient policy learning.
- Introduced advantage-weighted value guidance and achieved state-of-the-art results on D4RL, including perfect scores on multiple AntMaze tasks.

- Formulated sequential stochastic combinatorial optimization as a new decision-making setting with uncertain dynamics and combinatorial action spaces.
- Developed a hierarchical RL framework with tailored state, reward, and null-action design for efficient learning over large combinatorial action spaces.
- Proposed a wake-sleep option training strategy and outperformed strong baselines by 20% on adaptive influence maximization and route planning.

Publications

- [1] H. Meng, **X. Feng**, et al. “Solving Constrained Optimization Problems as ODE-based Models Using Reinforcement Learning,” *TMLR*, 2026.
- [2] **X. Feng**, L. Tang, C. Wang, and H. Chen. “Offline Reinforcement Learning with Generative Trajectory Policies,” *ICML*, 2026. *arXiv preprint* arXiv:2510.11499, 2025.
- [3] **X. Feng**, Z. Yu, Y. Xiong, and H. Chen. “Sequential Stochastic Combinatorial Optimization Using Hierarchical Reinforcement Learning,” *ICLR*, 2025. *arXiv preprint* arXiv:2502.05537, 2025.
- [4] B. Zhou, X. Guo, **X. Feng**, et al. “Complex-Frequency Selective Resonances in Disordered Time-Varying Cavities,” *Laser & Photonics Reviews*, e01228, 2025.
- [5] **X. Feng**, I. Roberts. “Adaptive Cell Range Expansion in Multi-Band UAV Communication Networks,” *arXiv preprint* arXiv:2411.18123, 2024.
- [6] B. Zhou, **X. Feng**, et al. “Transition from normal to non-normal distributions of an electromagnetic field in a disordered time-varying cavity,” *Physical Review A*, 110(6), 063524, 2024.
- [7] B. Zhou, **X. Feng**, et al. “Susceptibility Invariance and Duality-Matching Condition for Perpendicular-Motion Metasurface,” *Advanced Physics Research*, 3(10), 2400073, 2024.
- [8] Y. Wei, L. Liang, B. Zhou, and **X. Feng**. “A Modified Blockchain DPoS Consensus Algorithm Based on Anomaly Detection and Reward-Punishment,” *IEEE International Conference on Communication Software and Networks*, 283–288, 2021.

Teaching Experience

Graduate Teaching Assistant, *University of California, Los Angeles*

Spring 2024

Course: *Introduction to Communication Systems (ECE 132A)*. Graded assignments and exams, provided feedback to improve students’ understanding, and worked with the instructor to maintain consistent grading standards.

Undergraduate Teaching Assistant, *Chongqing University*

Fall 2021–Fall 2022

Courses: *Digital Signal Processing* (Fall 2021, Fall 2022); *Introduction to Programming with C++* (Fall 2022). Prepared course materials, supported lectures and office hours, graded homework and final exams, and was recognized as a “Top 10 Outstanding Teaching Assistant.”

Selected Honors and Awards

Distinguished Student, Chongqing University (top 10/266 in department)

May 2023

Top 10 Outstanding Teaching Assistant, Chongqing University

Jan. 2023

National Scholarship for Academic Excellence (top 4/266 in department)

Dec. 2022

Outstanding Graduate, Chongqing University (top 25/266 in department)

Nov. 2022

First Prize, Chinese Mathematics Competitions (twice; top 0.2% nationally)

Dec. 2020, Dec. 2021

Second Prize, Undergraduate Mathematical Contest in Modeling (top 2.3% nationally)

Nov. 2021

Professional Service

Reviewer, NeurIPS 2026

Jun. 2026

Reviewer, AISTATS 2026

Oct. 2025

Reviewer, IEEE GLOBECOM 2024 (Signal Processing for Communications Symposium)

Jun. 2024

Skills

Programming: Python, C++, Java, MATLAB

ML / Deep Learning: PyTorch, Hugging Face Transformers, Scikit-learn, NumPy, Pandas, OpenAI Gym

Generative AI & RL: Offline RL, Policy Optimization, Diffusion Models, LLM Fine-tuning, Model Evaluation

Systems & Tools: Linux, Git, Docker, Kubernetes